

Universal Futures Trading Bot V8

COMPLETE USER MANUAL

Multi-Exchange Futures • Strategy & Indicators • Grid Trading • Neutral Auto-Direction • Risk • SL/TP • Alerts
• Dashboard

Source: UniversalFuturesBot_V8_NEUTRAL_GRID_AUTO_DIRECTION_ALL_STRATEGIES_FINAL_AUDITED.py

Scope: This manual documents the behavior, controls, parameters, defaults, signal interpretation and execution flow implemented in the supplied script.

Supported exchanges in the source: Bybit, Binance, Gate.io, Bitget and WEEEX. KuCoin is not included.

Important: This is a software operation manual, not a profitability guarantee or trading recommendation. Futures and leverage can cause rapid losses. Test configuration and execution in demo/testnet before using live funds.

1. Manual at a Glance

Section	What it covers
2	Bot architecture and execution model
3	Connection & Market settings
4	Section 3 — Strategy & Indicators
5	Signal modes and voting
6	Section 4 — Grid Trading Engine
7	NEUTRAL_GRID automatic direction
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The manual follows the GUI organization in the program: Connection → Market → Strategy & Indicators → Grid Trading → Risk & SL/TP → Alerts & Dashboard.

2. Bot Architecture and Core Execution Model

The program is a Tkinter Windows GUI application. It maintains exchange state, strategy state, normal-position protection state, Grid state, performance statistics and CSV logging. Configuration is stored beside the executable/script in **config_universal_fixed.json**; trade records are stored in **universal_trade_logs_fixed.csv**.

The source explicitly separates the normal Strategy execution path from the Grid engine. When LONG_GRID, SHORT_GRID or NEUTRAL_GRID is selected, Grid owns execution; a Grid stop/cooldown does not fall through into the normal Direct Shot/SCORE entry engine.

- **Completed-candle logic:** strategy decisions use the last completed candle (normally index -2), reducing look-ahead from the currently forming candle.
- **Actual-entry protection:** normal SL/TP calculations use the actual filled position entry price and actual position quantity.
- **Protection verification:** protective orders are checked after creation; the code attempts to close a newly opened position if protection cannot be installed.
- **Old-order cleanup:** old open orders are cancelled before a new/reversed normal entry.
- **Grid protection:** Grid maintains a basket TP and global Grid SL and verifies that both remain active.
- **Global safety:** Max Daily Drawdown and Emergency Capital Loss Stop apply to both normal strategy and Grid.

The source comments state that the Grid Score is not a reduced Grid-only indicator list. It reuses the complete Section 3 directional module list and the same indicator settings/entry-mode logic.

3. Tab 1 — Connection & API Credentials

Setting	Default / options	Meaning
Exchange	bybit; binance; gate; bitget; weex	Selects the exchange connector.
API Key	blank	Exchange API credential.
API Secret	blank	Exchange API secret.
Account Mode	BYBIT_DEMO	BYBIT_DEMO, BYBIT_TESTNET, DEMO, TESTNET or LIVE. Exchange support differs.

The GUI notes: Bybit supports BYBIT_DEMO / BYBIT_TESTNET; Binance/Gate use TESTNET where supported; Bitget/WEEX use DEMO. Use credentials appropriate to the selected account mode.

Never paste real API credentials into chat. The script stores configuration in its JSON configuration file, so protect that file and use API keys with the minimum permissions required.

3.1 Tab 2 — Market & Execution

Setting	Default / options	Meaning
Symbol	BTC/USDT	Trading symbol passed to the exchange.
Timeframe	15m	Strategy candle timeframe. Options: 1m, 3m, 5m, 15m, 30m, 45m, 1h, 4h.
Leverage	5	Requested futures leverage.
Max Trades	10	Maximum completed trades for the session; 0 = unlimited.
No Re-Entry Same Candle	ON	Prevents instant re-entry after SL/TP/reversal on the same candle.
Cooldown (min)	0	Additional cooldown; 0 = OFF.
After SL Opposite Lock	ON	After SL/BE stop, blocks same-direction re-entry until a valid opposite signal.

The bot displays an estimated execution window based on timeframe/session settings, but actual duration can be longer if valid signals do not occur every candle.

4. Tab 3 — Strategy & Indicators

Section 3 contains the directional modules used by the normal Strategy engine and, importantly, by Grid SCORE and NEUTRAL_GRID when its direction source is SCORE/OFF. The complete module set is:

#	Module	Default enabled?
1	Supertrend	ON
2	EMA Filter	ON
3	EMA 9/20 Crossover	OFF
4	MACD Crossover	OFF
5	RSI	OFF
6	Bollinger Breakout	OFF
7	Stochastic Crossover	OFF
8	VWAP Trend Filter	OFF
9	VWAP Delta	OFF
10	Volumatic VIDYA	OFF
11	Nadaraya-Watson Envelope (NWE)	OFF
12	Liquidity Swings	OFF
13	Trendline Breakout	OFF
14	Multi-TF 4h Confluence	ON
15	Volume Filter	ON
16	ADX Filter	ON
17	ATR Volatility Filter	OFF

4.1 Supertrend

Supertrend is the main trend-direction module. The implementation follows a TradingView/Kivanc-style recurrence. ATR can use Wilder/RMA when the Change ATR Calculation Method option is ON, or SMA(True Range) when OFF.

Parameter	Default	Meaning
Enable	ON	Adds ST to the directional modules.
ATR Period	10	ATR calculation period.
ATR Mult	2.0	Distance multiplier around the source.
Source	CLOSE	CLOSE or HL2.
Change ATR Method	ON	ON = RMA/Wilder ATR; OFF = SMA of true range.
ST Entry	FRESH_FLIP	FRESH_FLIP uses a new trend flip; CURRENT_TREND follows the current ST direction.

For normal strategy voting, FRESH_FLIP is used for the ST vote except in SCORE mode, where the current ST trend is used. CURRENT_TREND always uses the current ST trend.

4.2 EMA Filter

The EMA module compares the completed candle close with the EMA.

Parameter	Default	Meaning
Enable	ON	Adds EMA to directional modules.
EMA Period	200	EMA length.
BUY direction	Close > EMA	Completed candle closes above EMA.
SELL direction	Close < EMA	Completed candle closes below EMA.

4.3 EMA 9/20 Crossover

Optional fast/slow EMA crossover module.

Parameter	Default	Meaning
Enable	OFF	Enable EMA crossover voting.
Fast	9	Fast EMA period.
Slow	20	Slow EMA period.
Entry	FRESH_CROSS	FRESH_CROSS requires a new crossover; CURRENT_TREND follows fast EMA above/below slow EMA.

4.4 MACD

MACD uses fast EMA minus slow EMA, with a signal EMA. The normal directional event is a MACD/signal crossover.

Parameter	Default	Meaning
Enable	OFF	Enable MACD voting.
Fast	12	Fast EMA.
Slow	26	Slow EMA.
Signal	9	Signal EMA.
BUY	Fresh bullish cross	MACD crosses above its signal on the completed candle.
SELL	Fresh bearish cross	MACD crosses below its signal on the completed candle.

4.5 RSI

RSI is Wilder-style RSI. The GUI supports reversal-zone logic and RSI/MA crossover logic.

Parameter	Default	Meaning
Enable	OFF	Enable RSI voting.
Period	14	RSI length.
OB	80	Overbought threshold.
OS	20	Oversold threshold.
RSI Logic	REVERSAL_ZONE	REVERSAL_ZONE, CROSS_MA or EITHER.
MA Type	EMA	SMA, EMA or WMA for the RSI MA.
MA Period	9	RSI MA length.

REVERSAL_ZONE: BUY when RSI is at/below OS and SELL when RSI is at/above OB. CROSS_MA: BUY when RSI crosses above its MA and SELL when it crosses below. EITHER accepts either event. The code treats flat/undefined cases conservatively; RSI is 50 in a flat market and 100 when average loss is zero with positive gain.

4.6 Bollinger Breakout

Parameter	Default	Meaning
Enable	OFF	Enable BB voting.
Period	20	Middle-band SMA period.

Parameter	Default	Meaning
StdDev	2	Standard-deviation multiplier.
BUY	Close > middle OR upper	Directional interpretation in the strategy.
SELL	Close < middle OR lower	Directional interpretation in the strategy.

4.7 Stochastic

Parameter	Default	Meaning
Enable	OFF	Enable Stochastic voting.
K	14	Raw %K lookback.
K Smooth	3	Smoothing applied to %K.
D	3	Moving average period for %D.
BUY	K crosses above D	Completed-candle bullish crossover.
SELL	K crosses below D	Completed-candle bearish crossover.

4.8 VWAP

Parameter	Default	Meaning
Enable	OFF	Enable rolling VWAP voting.
Period	50	Rolling volume-weighted period.
BUY	Close > VWAP	Completed candle closes above VWAP.
SELL	Close < VWAP	Completed candle closes below VWAP.

4.9 VWAP Delta

VWAP Delta measures price relative to session VWAP and can be smoothed with an HMA. The baseline is an EMA of the delta.

Parameter	Default	Meaning
Enable	OFF	Enable VWAP Delta.
Smooth	OFF	Enable HMA smoothing.
HMA	21	Smoothing length.
Baseline	50	EMA baseline length.
Logic	CURRENT_TREND	CURRENT_TREND uses delta vs baseline; CROSS_BASELINE requires a fresh crossing.

4.10 Volumatic VIDYA

The implementation calculates VIDYA using momentum-derived CMO weighting, an ATR-based band and a smoothed VIDYA line. It also tracks up/down volume and a volume delta percentage.

Parameter	Default	Meaning
Enable	OFF	Enable VIDYA voting.
Length	10	VIDYA base length.
Momentum	20	Momentum window used for CMO weighting.
Band	2	ATR band distance multiplier.
Entry	CURRENT_TREND	CURRENT_TREND follows VIDYA trend; FRESH_FLIP requires a new VIDYA direction change.

4.11 Nadaraya-Watson Envelope (NWE)

The source uses a causal/end-point Nadaraya-Watson calculation. The UI exposes a repaint checkbox, but it is fixed OFF: the bot is explicitly non-repainting and uses completed candles and past data.

Parameter	Default	Meaning
Enable	OFF	Enable NWE voting.
Bandwidth	8	Kernel bandwidth.

Parameter	Default	Meaning
Multiplier	3	MAE envelope multiplier.
Entry	FRESH_CROSS	FRESH_CROSS uses close crossing the envelope; CURRENT_TREND uses the slope of the NWE center.
Repaint	OFF / fixed	Not user-configurable in the trading calculation; NWE is always causal/non-repainting.

4.12 Liquidity Swings

Liquidity Swings uses confirmed swing highs/lows and price breaks. A pivot is only known after the configured number of candles have closed to its right, so the strategy does not use a future pivot before it is confirmed.

Parameter	Default	Meaning
Enable	OFF	Enable liquidity-swing voting.
Pivot Lookback	14	Pivot confirmation length.
Swing Area	Wick Extremity	Wick Extremity or Full Range.
Filter	Count	Count or Volume.
Filter Value	0	Minimum count/volume condition for a break.
Entry	FRESH_BREAK	FRESH_BREAK uses a new liquidity break; CURRENT_TREND stays with the last breakout direction.

4.13 Trendline Breakout

Trendlines are built from confirmed pivot highs/lows. Resistance must slope down; support must slope up. Breakouts use completed candles.

Parameter	Default	Meaning
Enable	OFF	Enable trendline voting.
Pivot Lookback	14	Pivot confirmation length.
Min Pivot Distance	5	Minimum spacing between accepted pivots.
Mode	FRESH_BREAK	FRESH_BREAK, CURRENT_TREND or BREAK_RETEST.
Breakout Buffer %	0	Extra percentage buffer beyond trendline.
Retest Candles	3	Maximum retest window.

FRESH_BREAK = new crossing. CURRENT_TREND = remain with the last breakout direction. BREAK_RETEST = require the breakout plus a qualifying retest within the configured window.

4.14 Multi-TF (4h Confluence)

When enabled, the bot fetches 4-hour OHLCV and calculates a 200-period EMA. It uses the last completed 4h candle.

Parameter	Default	Meaning
Enable	ON	Enable 4h confluence.
4h BUY	4h close > 4h EMA200	Bullish 4h filter.
4h SELL	4h close < 4h EMA200	Bearish 4h filter.

4.15 Volume Filter

Volume is a participation/strength condition. The code calculates a volume moving average and, when the volume condition passes, gives the directional vote of the completed candle.

Parameter	Default	Meaning
Enable	ON	Enable Volume module.
Volume MA Length	20	Moving-average period.
Pass condition	Volume >= Volume MA	Only then Volume enters the directional module list.
Direction	Candle body	Close > Open = BUY; Close < Open = SELL.

4.16 ADX Filter

ADX is calculated with Wilder/RMA smoothing. ADX first has to meet the threshold; direction is then taken from +DI versus -DI.

Parameter	Default	Meaning
Enable	ON	Enable ADX module.
ADX Threshold	20	Minimum ADX required for the module to participate.
BUY	+DI > -DI	Only after ADX >= threshold.
SELL	-DI > +DI	Only after ADX >= threshold.

4.17 ATR Volatility Filter

ATR is a volatility condition, not inherently directional. When ATR percentage meets the minimum threshold, its directional vote follows the completed candle.

Parameter	Default	Meaning
Enable	OFF	Enable ATR module.
Min ATR %	0.30	Minimum ATR as a percentage of price.
Direction	Candle body	Close > Open = BUY; Close < Open = SELL.

5. Signal Modes and Voting System

Mode	Minimum / rule	Meaning
SINGLE_SIGNAL	1	One enabled directional signal is enough. It does not require a second indicator, Volume, ADX, ATR or MTF agreement.
2_SIGNALS	2	At least two aligned directional votes are required.
3_SIGNALS	3	At least three aligned votes are required.
4_SIGNALS	4	At least four aligned votes are required.
SCORE	User Score setting; default 4	Counts BUY and SELL votes. Direction is selected when the required score is reached with directional dominance.
STRICT_ALL_FILTERS	All active directional modules	Requires all active directional conditions to agree for the selected direction.

The program builds one **directional_modules** list from the enabled Section 3 indicators. Each module contributes a BUY vote, a SELL vote, or neither, depending on its completed-candle state and its configured entry mode.

For SCORE, the GUI field **Score (SCORE mode)** defaults to 4. Grid Score Min is a separate threshold and defaults to 1.

The same completed-candle directional module list is copied for Grid scoring. Therefore Grid SCORE uses the same indicator enable/disable switches and indicator parameters as Section 3.

6. Tab 4 — Grid Trading Engine

Grid has its own order sizing, spacing, basket TP, global Grid SL, exposure limit, Grid drawdown limit, trend filter, score threshold, recentering and cooldown controls. Normal Strategy risk-per-trade/fixed quantity and normal Strategy SL/TP are not used for Grid entries.

Setting	Default	Meaning
Grid Mode	OFF	OFF, DIRECT_SHOT, LONG_GRID, SHORT_GRID, NEUTRAL_GRID.
Grid Levels	5	Number of planned grid levels.
Spacing %	1.0	Distance of each level from center.
Order Size USDT	10	Base notional size per Grid order.
Size Increase %	0	Compounds order notional by level.
Grid TP %	1.0	Basket TP measured from actual average Grid position entry.
Global Grid SL %	5.0	Grid basket stop measured from Grid center.
Max Exposure USDT	100	Maximum current + planned Grid exposure.
Max Grid DD %	3.0	Grid-session/equity drawdown stop.
Trend Filter	OFF	OFF, SUPERTREND or SCORE.
Grid Score Min	1	Minimum directional score for Grid SCORE / Neutral auto-direction.
Recenter	OFF	If enabled and flat, recenters after price moves the configured distance.

Setting	Default	Meaning
Recenter Distance %	3.0	Distance from center required to recenter.
Cooldown After Grid Stop	30 min	Grid pause after a Grid stop.

6.1 LONG_GRID

LONG_GRID is fixed-direction. It places BUY limit orders below the center. The existing script does not automatically change LONG_GRID into SHORT_GRID.

6.2 SHORT_GRID

SHORT_GRID is fixed-direction. It places SELL limit orders above the center. It does not automatically change into LONG_GRID.

6.3 DIRECT_SHOT

DIRECT_SHOT leaves execution with the normal strategy path. The Grid manager returns control to the normal entry engine when Grid Mode is OFF or DIRECT_SHOT.

6.4 NEUTRAL_GRID — Automatic Direction

NEUTRAL_GRID is the only Grid mode that automatically changes between LONG and SHORT based on the selected directional source.

Trend Filter	NEUTRAL_GRID direction source
SUPERTREND	Follows the current Supertrend direction: bullish → LONG; bearish → SHORT.
SCORE	Uses the complete Section 3 directional-module score. BUY score must reach Grid Score Min and be greater than SELL score for LONG; vice versa for SHORT.
OFF	In this version, Neutral auto-direction also uses the complete Section 3 score logic. A tie or insufficient score produces no new Grid entries.

7. NEUTRAL_GRID Automatic Direction — Exact Runtime Behavior

The automatic direction state can be LONG, SHORT or None. When the direction changes, the bot performs a safety transition rather than simply placing opposite orders on top of existing inventory.

Example: Supertrend BUY → LONG Grid

The bot selects LONG, sets the Grid center, cancels disallowed entries, and places BUY limit levels below the center.

Example: Supertrend changes BUY → SELL

- Pending BUY Grid entry orders are cancelled first.
- If an existing Grid position is LONG, the bot closes that inventory at market.
- The bot waits and checks that the position is actually flat.
- Only after flat confirmation does it remove old Grid TP/SL protection.
- The Grid state is reset and the center is reset to the current market price.
- The auto-direction becomes SHORT and SELL Grid levels are placed above the new center.

Temporary neutral/tie

If the score has no valid dominant direction, pending entries are cancelled and no new entries are placed. An already-open protected position is not forcibly closed merely because the indicators temporarily tie/turn neutral.

Same direction returns

If the existing position is already on the newly selected side, the bot retains that position and its protection rather than unnecessarily closing and reopening it.

If the bot cannot verify that a reversal position has become flat, it stops the Grid rather than starting the opposite side on uncertain inventory. This is an intentional fail-safe in the supplied code.

7.1 Grid Price and Size Calculation

For LONG-side Grid levels, the code places BUY limits below center:

$$\text{Level } n \text{ price} = \text{center} \times (1 - \text{spacing} \times n)$$

For SHORT-side Grid levels, it places SELL limits above center:

$$\text{Level } n \text{ price} = \text{center} \times (1 + \text{spacing} \times n)$$

Base order notional is increased by the configured Size Increase percentage:

$$\text{Level } n \text{ USDT} = \text{Order Size} \times (1 + \text{Size Increase})^{(n-1)}$$

The Grid engine checks planned exposure plus current position exposure against Max Exposure before placing another level.

The source validates Global Grid SL against the total Grid depth: Global Grid SL must be greater than Spacing × Levels, so the outermost Grid level does not coincide with the SL boundary.

8. Tab 5 — Dynamic Risk & Sizing

The sizing controls in this section are explicitly labelled **NORMAL STRATEGY ONLY**. They do not size Grid orders.

Setting	Default / options	Meaning
Sizing Mode	EQUITY_RISK_%	EQUITY_RISK_% or FIXED_QTY for normal Strategy entries.
Risk Per Trade %	1.0	Normal Strategy risk-based sizing input.
Fixed Qty	0.001	Normal Strategy fixed quantity when FIXED_QTY is selected.
Max Daily Drawdown %	5.0	Global safety stop for both normal Strategy and Grid.
Emergency Capital Loss Stop %	30.0	Equity loss from bot start; closes all account positions and stops bot.

Grid has independent order size and exposure controls. However, the global Max Daily Drawdown and Emergency Capital Loss Stop remain applicable to Grid.

9. Tab 5 — Normal Strategy Stop Loss & Take Profit

This section is explicitly labelled **NORMAL STRATEGY SL/TP ONLY — NOT USED BY GRID**.

Setting	Default / options	Meaning
SL Mode	PRICE_%	PRICE_% or ROI_%.
TP Mode	ROI_%	PRICE_% or ROI_%.
SL Target	1.5	Normal Strategy SL target.
TP1 Target	2.0	Normal Strategy first target.
TP2 Target	4.0	Normal Strategy second target.
Move SL to Break-Even after TP1	ON	Moves normal Strategy protection to break-even after TP1.
TP Close Qty Mode	PERCENT_%	PERCENT_% or FIXED_QTY.
TP1 Close	50	50% in percentage mode or 50 units in fixed-quantity mode.
TP2 Close	50	Second portion; percentage mode is intended to total 100%.
Hold-All-Reverse SL ROI	5.0	Threshold used when Hold-All-Reverse behavior is enabled.
Wait for ALL active signals to reverse	OFF	When ON with Hold-All-Reverse, threshold is monitored by bot and position waits for all active directional modules to reverse.

The source explains the two target modes as: **PRICE_%** = market-price movement, while **ROI_%** = position ROI target converted to an exchange price trigger using leverage. Example in the source: at 20×, 20% ROI is approximately a 1% price move.

Normal SL/TP protection is based on the actual filled entry price and actual position quantity. The bot uses exchange-specific trigger creation for supported exchanges and verifies protection after creation.

9.1 Hold-All-Reverse Behavior

- **OFF**: normal exchange SL can close the position at the configured threshold.
- **ON**: the configured threshold is monitored by the bot; the threshold is not used as a normal exchange hard stop in this special mode.

- If the additional 'WAIT for ALL active signals to reverse' control is ON, the bot waits for every active directional module to reverse before exiting by strategy reversal.

Hold-All-Reverse can deliberately keep a position open longer than a conventional exchange stop. Understand this behavior before enabling it.

10. Tab 6 — Alerts & Dashboard

Setting	Default	Meaning
Enable Telegram Alerts	OFF	Turns Telegram notifications on/off.
Bot Token	blank	Telegram bot token.
Chat ID	blank	Destination chat ID.

The dashboard displays session statistics including starting/current balance, net PnL, completed trades, wins, losses and win rate. The execution log records runtime events and Grid/strategy decisions.

CSV trade logging uses fields including Timestamp, Exchange, Symbol, Side, ActualEntry, Qty, SL, TP1, TP2 and Notes.

11. Startup, Runtime and Shutdown

11.1 Starting the Bot

- Select exchange and account mode.
- Enter symbol and timeframe.
- Set leverage appropriate to the test account.
- Configure Section 3 indicators and signal mode.
- If using Grid, validate Grid Mode and Grid parameters.
- Configure global safety limits.
- Save configuration if desired, then Start Bot.
- The bot logs connection, balance, SL/TP mode, session limits, enabled modules and Grid settings.

11.2 Runtime cycle

Stage	What the engine does
Market data	Fetches OHLCV for the selected strategy timeframe.
Completed candle	Uses the last completed candle for strategy decisions.
Indicators	Calculates enabled indicators and filters.
Directional modules	Builds the common list of enabled directional votes.
Scores	Calculates normal BUY/SELL score and the same Grid BUY/SELL score.
Grid branch	If Grid mode is active, Grid manages entries/protection and owns execution.
Normal branch	If Grid does not own execution, normal signal/position logic manages entries/reversals.
Protection	Checks/reconciles SL/TP protection as required.
Safety	Checks drawdown, exposure, emergency stop and re-entry gates.
Logging	Writes GUI logs, CSV trade data and optional Telegram alerts.

11.3 Stopping the Bot

Manual stop and GUI shutdown explicitly stop active Grid modes, close Grid inventory through the Grid stop path, and clean up managed Grid orders subject to the safety checks in the source.

12. Complete Parameter Reference — Quick Lookup

Connection	Exchange	bybit	Connector selection
Connection	Account Mode	BYBIT_DEMO	Demo/testnet/live selection
Market	Symbol	BTC/USDT	Trading symbol

Connection	Exchange	bybit	Connector selection
Market	Timeframe	15m	Strategy candle timeframe
Market	Leverage	5	Requested leverage
Market	Max Trades	10	0 = unlimited
Market	No Re-Entry Same Candle	ON	Same-candle re-entry gate
Market	Cooldown	0 min	0 = OFF
Strategy	Supertrend	ON	Main trend module
Strategy	ST ATR Period	10	Supertrend ATR period
Strategy	ST ATR Mult	2.0	Supertrend multiplier
Strategy	ST Source	CLOSE	CLOSE / HL2
Strategy	ST ATR Method	ON	ON RMA / OFF SMA(TR)
Strategy	ST Entry	FRESH_FLIP	FRESH_FLIP / CURRENT_TREND
Strategy	EMA	ON	Close vs EMA
Strategy	EMA Period	200	EMA length
Strategy	EMA Cross	OFF	Fast/slow crossover
Strategy	EMA Fast/Slow	9 / 20	Crossover periods
Strategy	EMA Cross Entry	FRESH_CROSS	Fresh cross / current trend
Strategy	MACD	OFF	12 / 26 / 9
Strategy	RSI	OFF	14 / OB 80 / OS 20
Strategy	RSI Logic	REVERSAL_ZONE	REVERSAL_ZONE / CROSS_MA / EITHER
Strategy	RSI MA	EMA / 9	RSI moving average
Strategy	Bollinger	OFF	20 / 2
Strategy	Stochastic	OFF	14 / 3 / 3
Strategy	VWAP	OFF	Rolling period 50
Strategy	VWAP Delta	OFF	Baseline 50; HMA 21 optional
Strategy	VIDYA	OFF	10 / 20 / band 2
Strategy	NWE	OFF	Bandwidth 8 / multiplier 3
Strategy	Liquidity Swings	OFF	Pivot 14 / Count / value 0
Strategy	Trendline	OFF	Pivot 14 / distance 5 / buffer 0 / retest 3
Strategy	MTF	ON	4h close vs EMA200
Strategy	Volume	ON	Volume MA 20
Strategy	ADX	ON	Threshold 20
Strategy	ATR	OFF	Minimum ATR 0.30%
Strategy	Signal Mode	SINGLE_SIGNAL	Six available modes
Strategy	Score	4	Normal SCORE threshold
Strategy	Hold Until All Reverse	ON	Normal position reversal-hold rule
Grid	Grid Mode	OFF	OFF / DIRECT_SHOT / LONG / SHORT / NEUTRAL
Grid	Levels	5	Grid levels
Grid	Spacing	1.0%	Distance between levels
Grid	Order Size	10 USDT	Base Grid notional
Grid	Size Increase	0%	Per-level notional increase
Grid	Grid TP	1.0%	Basket TP
Grid	Global Grid SL	5.0%	Basket SL
Grid	Max Exposure	100 USDT	Exposure cap
Grid	Max Grid DD	3.0%	Grid DD stop
Grid	Trend Filter	OFF	OFF / SUPERTREND / SCORE
Grid	Grid Score Min	1	Grid directional threshold
Grid	Recenter	OFF	Flat-grid recentering
Grid	Recenter Distance	3.0%	Recenter trigger
Grid	Cooldown	30 min	Grid stop cooldown
Risk	Sizing Mode	EQUITY_RISK_%	Normal Strategy only

Connection	Exchange	bybit	Connector selection
Risk	Risk Per Trade	1.0%	Normal Strategy only
Risk	Fixed Qty	0.001	Normal Strategy only
Risk	Max Daily DD	5.0%	Global
Risk	Emergency Loss Stop	30.0%	Global account safety
SL/TP	SL Mode	PRICE_%	Normal Strategy only
SL/TP	TP Mode	ROI_%	Normal Strategy only
SL/TP	SL	1.5%	Normal Strategy target
SL/TP	TP1	2.0%	Normal Strategy target
SL/TP	TP2	4.0%	Normal Strategy target
SL/TP	TP1 BE	ON	Normal Strategy
SL/TP	TP Qty Mode	PERCENT_%	PERCENT_% / FIXED_QTY
SL/TP	TP1 Close	50	Half in percent mode
SL/TP	TP2 Close	50	Half in percent mode
SL/TP	Hold-All-Reverse ROI	5.0%	Special normal Strategy behavior
SL/TP	Wait All Reverse	OFF	Special normal Strategy behavior
Alerts	Telegram	OFF	Notifications

13. Practical Setup and Testing Checklist

The following is an operational checklist, not a claim that any parameter combination is profitable.

- **Step 1 — Demo first:** use the exchange's demo/testnet mode and verify the correct symbol and account.
- **Step 2 — Start simple:** enable only a small number of indicators until you understand each vote.
- **Step 3 — Inspect logs:** confirm the log shows the expected signal mode, enabled modules, Grid mode and scores.
- **Step 4 — Grid test:** start with small Grid Order Size and a low Max Exposure so execution behavior can be observed.
- **Step 5 — Neutral test:** use NEUTRAL_GRID and watch a controlled Supertrend direction change. Confirm old pending orders are cancelled and the direction transition is logged.
- **Step 6 — SCORE test:** enable multiple Section 3 modules and confirm Grid logs show the same strategy module names and B/S score.
- **Step 7 — Protection test:** verify that Grid TP and Global Grid SL appear as exchange-side orders and remain present after position changes.
- **Step 8 — Safety test:** test stop/cooldown, maximum exposure and drawdown behavior in demo.
- **Step 9 — Only then consider live operation:** re-check API permissions, symbol precision, leverage and risk limits.

A successful software test does not guarantee exchange execution under every network, liquidity, API, symbol or account-mode condition. The source itself treats protection verification and exchange reads as important safety gates.

14. Troubleshooting

Symptom	What to inspect
No Grid orders	Grid Mode, Trend Filter, current direction, score threshold, Max Exposure, existing open orders and Grid cooldown.
NEUTRAL_GRID does not choose a side	SUPERTREND may be neutral/unavailable, or SCORE/OFF may have a tie or fail Grid Score Min.
BUY Grid disappears after signal change	Expected when current filter blocks BUY; pending disallowed entries are cancelled.
Grid does not immediately reverse	The bot first cancels pending entries, closes existing opposite inventory and verifies flat before rebuilding the new side.
Grid stops unexpectedly	Check Grid DD, Max Exposure, protection verification, unexpected opposite position or exchange/order errors.
Normal strategy does not re-enter	Check No Re-Entry Same Candle, cooldown, post-SL opposite lock, signal mode and active filters.
NWE repaint option cannot be enabled	This is intentional in the supplied version: NWE is fixed to non-repainting/causal mode.
TP/SL appears different from candle close	Normal protection uses the actual filled position entry; Grid TP uses actual average Grid entry while Global Grid SL is based on Grid center.

15. Glossary

Term	Meaning in this bot
Completed candle	The last fully closed candle, generally referenced as index -2.
Directional module	An enabled strategy component that can provide BUY/SELL direction.
BUY score	Count of directional modules currently voting bullish.
SELL score	Count of directional modules currently voting bearish.
Grid center	Reference price from which Grid levels are calculated.
Grid basket TP	One reduce-only TP for the Grid position using the actual average entry.
Global Grid SL	Grid-wide protective stop based on the Grid center.
Recenter	Move the Grid center to current market price when flat and the distance threshold is reached.
Fresh event	A newly occurring crossover/flip/break rather than an already-established trend.
Current trend	Continue using the current established directional state.
ROI mode	Convert desired position ROI into an exchange price trigger using leverage.
Fail closed	Stop/flatten rather than continue trading when required protection cannot be verified.
One-way position behavior	The source's Grid logic expects a single net position direction on the symbol; opposite inventory is treated as an error for fixed-direction Grid modes.

16. Source-Implementation Notes

This manual is derived from the supplied Python source rather than from a generic futures-bot description. The source explicitly documents TradingView-based translations for Supertrend, VWAP Delta, Volumatic VIDYA, Nadaraya-Watson Envelope and Liquidity Swings, with visual-only drawing objects omitted where applicable.

The source states that NWE calculations are causal/end-point and non-repainting, Liquidity Swings uses confirmed pivots, Trendline Breakout uses confirmed pivots, and strategy decisions are based on completed candles.

The program also contains exchange-specific trigger creation for Bybit and Binance and a generic trigger path for other supported exchanges. The Grid protection engine installs SL first, creates basket TP, verifies both, and only then removes obsolete protection.

Version scope: This document describes the uploaded file supplied for this manual. If the Python file is modified later, this PDF should be regenerated so its parameter defaults and behavior remain synchronized with the code.